



NeuroSpace

Complete evaluation export

Job ID	3afc6cc1-01f2-49c7-8ca1-312f2e40fd0c
Status	completed
Result timestamp	2026-05-15T19:58:54.580484
PDF generated	2026-05-15 21:16:25

Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	30
manual_overrides.qrnn_epochs	10
manual_overrides.batch_size	32
manual_overrides.rnn_batch_size	32
manual_overrides.qrnn_batch_size	16
num_classes	2
class_names	0, 1
qrnn_backend	qiskit
noise_enabled	False
noise_level	0.0000
mitigation_enabled	False
mitigation_runs	1

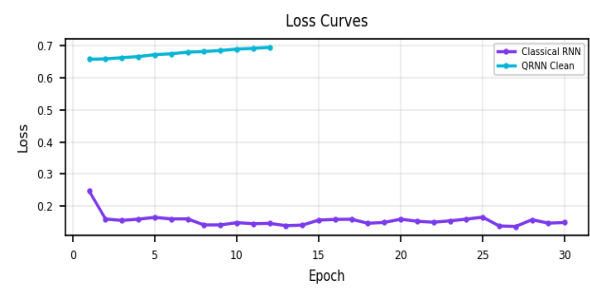
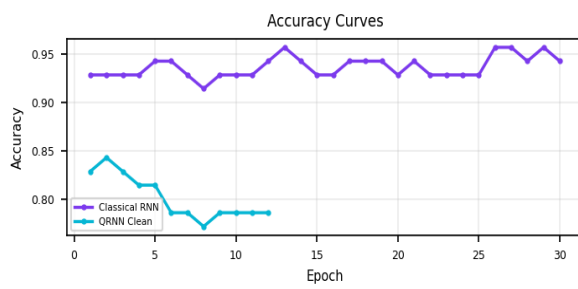
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	93.57%	79.29%	14.29%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	93.57%	93.55%	93.57%	93.55%	0.1485
QRNN Clean	79.29%	80.38%	79.29%	77.45%	0.6957

Learning Curves



Classical RNN

Metric	Value
Accuracy	93.57%
Precision	93.55%
Recall	93.57%
F1 Score	93.55%
Loss	0.1485
Train Loss	0.0461
Val Loss	0.1485
Train Accuracy	98.77%
Val Accuracy	94.29%
Train Size	489
Val Size	70
Test Size	140

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	5	43

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
0	94.62%	95.65%	95.14%	92.0000
1	91.49%	89.58%	90.53%	48.0000
macro avg	93.06%	92.62%	92.83%	140.0000
weighted avg	93.55%	93.57%	93.55%	140.0000
Accuracy			93.57%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.3434	0.2469	84.46%	92.86%
2	0.1566	0.1592	95.71%	92.86%
3	0.1213	0.1550	96.52%	92.86%
4	0.1040	0.1588	96.73%	92.86%
5	0.1392	0.1644	97.14%	94.29%
6	0.0997	0.1596	97.55%	94.29%
7	0.0779	0.1597	98.36%	92.86%
8	0.0779	0.1409	97.34%	91.43%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.0635	0.1408	98.16%	92.86%
10	0.0596	0.1478	98.16%	92.86%
11	0.0797	0.1448	97.34%	92.86%
12	0.0767	0.1457	97.55%	94.29%
13	0.0723	0.1385	97.34%	95.71%
14	0.0580	0.1402	98.16%	94.29%
15	0.0448	0.1559	98.77%	92.86%
16	0.0567	0.1581	97.75%	92.86%
17	0.0972	0.1586	97.75%	94.29%
18	0.0891	0.1459	97.14%	94.29%
19	0.0599	0.1488	98.16%	94.29%
20	0.0705	0.1588	98.77%	92.86%
21	0.0609	0.1523	98.16%	94.29%
22	0.0484	0.1493	98.98%	92.86%
23	0.0455	0.1536	98.57%	92.86%
24	0.0522	0.1590	98.36%	92.86%
25	0.0786	0.1650	98.16%	92.86%
26	0.0455	0.1376	98.57%	95.71%
27	0.0556	0.1360	97.96%	95.71%
28	0.0451	0.1572	98.77%	94.29%
29	0.0668	0.1464	97.75%	95.71%
30	0.0461	0.1485	98.77%	94.29%

QRNN Clean

Metric	Value
Accuracy	79.29%
Precision	80.38%
Recall	79.29%
F1 Score	77.45%
Loss	0.6957
Train Loss	0.6042
Val Loss	0.6957
Train Accuracy	86.30%
Val Accuracy	78.57%
Train Size	489
Val Size	70
Test Size	140

QRNN Clean Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	25	23

QRNN Clean Classification Report

Class	Precision	Recall	F1 Score	Support
0	77.88%	95.65%	85.85%	92.0000
1	85.19%	47.92%	61.33%	48.0000
macro avg	81.53%	71.78%	73.59%	140.0000
weighted avg	80.38%	79.29%	77.45%	140.0000
Accuracy			79.29%	

QRNN Clean Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6660	0.6585	84.05%	82.86%
2	0.6503	0.6596	83.84%	84.29%
3	0.6383	0.6635	86.30%	82.86%
4	0.6302	0.6669	85.48%	81.43%
5	0.6228	0.6728	85.28%	81.43%
6	0.6161	0.6757	85.48%	78.57%
7	0.6131	0.6808	85.69%	78.57%
8	0.6115	0.6827	86.91%	77.14%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.6087	0.6859	86.30%	78.57%
10	0.6094	0.6903	86.09%	78.57%
11	0.6062	0.6926	86.30%	78.57%
12	0.6042	0.6957	86.30%	78.57%